

AFL Data Foundations — Data Dictionary & Prediction Target Definitions

1983–2025 · 4 source tables · 7,904 matches · 217,568 player-games · 2,848 players · Contract for Day-2 modelling — every model this week should read targets and features exactly as defined here.

Source Tables & Grain

File	Grain (one row =)	Rows	Join keys
afl_players_info_raw.csv	one player (career record)	2,848	id → player_id
afl_players_seasonal_stats_raw.csv	one player–season–team–phase (home-and-away / finals)	25,491	player_id, year, team
round_by_round_stats.csv	one player-game	217,568	player_id; team+match_date → match
team_matches_home_away_raw.csv	one team-match (2 rows/match: H & A)	15,808	match_date + team_name (unique)

Prediction Targets

Target	Grain	Type	Formula
result	match	3-class classification	'H' / 'A' / 'D' from home_score vs away_score
home_win	match	Binary classification	1 if result='H', 0 if result='A', NaN if draw (draws ~0.8% of matches)
margin	match	Regression	home_score – away_score (signed)
is_match_top_disposals	player-game	Binary flag	disposals == max(disposals) within match_id (disposals recomputed = kicks+handballs)
is_match_top_goals	player-game	Binary flag	goals == max(goals) within match_id
Top disposal-getter (season)	player-season	Leaderboard	Rank by sum(disposals); min. games-played threshold recommended
Top goal-kicker (season)	player-season	Leaderboard	Rank by sum(goals) — Coleman Medal framing
Fantasy composite (season)	player-season	Leaderboard	Rank by sum(fantasy_points) — source-provided composite, kept opaque/as-is
Brownlow votes (season)	player-season	Leaderboard	Rank by sum(brownlow_votes) — real umpire-awarded best-and-fairest signal

Key Engineered Features (versioned feature table v1 — all leak-free: computed only from games strictly before the match being predicted)

Feature family	Computation window	Source columns
form_win_rate_last{3,5,10}, form_avg_score_for/against_last{N}, form_avg_margin_last{N}	trailing N games, shift(1) before rolling	team win / team_score / opp_score / margin
team_streak_entering_game	running win(+)/loss(–) streak, all prior games	team win
days_since_last_match, games_played_this_season, venue_experience	previous match / season-to-date / all prior history	match_date, season, venue
ladder_position_prior, percentage_prior	cumulative points & percentage, current game excluded	team win/draw, team_score, opp_score
h2h_win_rate_prior, h2h_games_played_prior	all-time prior meetings vs this specific opponent	team, opponent, win
player_{disposals,goals,fantasy_points,tackles,player_score}_avg_last{3,5,10}	trailing N games, shift(1) before rolling	round_by_round_stats.csv player columns
player_games_played_prior, player_career_avg_disposals_prior	career-to-date, current game excluded	player_id, match_date

Data-quality fixes applied (see notebook Task 1.4 for full detail): team names canonicalised across 3 casings + tab/abbrev. variants (20 clubs); 5 duplicate player rows & 10 duplicate season rows dropped; round_by_round 'score' column (always 0 / unusable) dropped; player_score = goals×6+behinds derived instead; 'disposals' recomputed as kicks+handballs (raw column disagreed with kicks+handballs for 9.7% of rows, incl. 720 impossible negative values); one match's margin/result corrected from its own reported scores. **Split:** strict time-based split (train on earlier seasons, hold out most recent season) via afl_lib.time_based_split() — a random row split would leak season-level outcome information into training. **Realistic ceiling:** ~65–72% accuracy on binary match-winner is a reasonable target; near-100% signals leakage, not skill.